Capstone Design Project — Final Report

The Effect of Reward Function Design in RL-Based BTC Grid Trading: Pareto Frontier, Winner Reversal, and Out-of-Distribution Robustness of Prospect-Theoretic Reward

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Abstract

We quantitatively study the effect of *reward function design* on PPO-based reinforcement-learning policies for ATR-proportional grid trading on the BTC/USDT 1-hour market. Four reward variants—symmetric (`sym`), asymmetric ($\beta = 3.42$, `asym`), Differential Sharpe Ratio ($\eta = 1/28\,\mathrm{h}$, `dsr`; Moody and Saffell 2001), and prospect-theoretic ($\alpha = 0.68$, $\lambda = 3.30$, `pt`; Kahneman and Tversky 1979)—are each trained with 10 seeds × 1M steps and evaluated on three environments: (i) single-split validation (Val 2021–2023), (ii) 6-fold combinatorial purged cross-validation [López de Prado, 2018] 15 paths, and (iii) the unsealed out-of-sample Test set (2024–2026, BTC \$42K → \$76K bull regime).

We report five key findings. **(1)** All four variants exceed the ATR baseline by +11–24% on Val, yet partition into two clusters {`sym`, `dsr`} (aggressive) vs. {`asym`, `pt`} (conservative); within-cluster Cohen's $|d| < 0.3$ versus across-cluster $|d| > 0.8$. **(2)** Mechanism: reward loss asymmetry directly determines trade frequency and holding time (policy distance ratio 2.22×), and the sliding-window memory structure of DSR results in 2–6× higher hold rate than other variants. **(3)** Winner reverses across evaluation environments: Val=`sym`(1.87) → CPCV=`dsr`(1.41, $p < 0.001$) → Test=`pt`(0.34–0.37, consistent across two model sources, $p < 0.002$). **(4)** The OOS robustness of `pt` arises from a short-holding mechanism: loss aversion $\lambda = 3.30$ combined with concave gain $\alpha = 0.68$ trains policies to exit within mean 1.4 h (max 6 h), thereby avoiding the sell-side timing risk of the unseen bull regime. In contrast, `dsr` learns long holding (mean 4.58 h, max 169 h = 7 days) and records the worst OOS performance—demonstrating that the same reward formulation underlies both in-sample advantage and OOS failure as two sides of the same coin. **(5)** The Val–Test distribution shift is highly significant (KS $p < 10^{-10}$, −27% variance), while policies' actions remain nearly invariant between Val and Test ($\Delta \leq 5\%$), indicating that the regime shift, not the policy, drives OOS differences.

Our contribution is twofold. First, we demonstrate quantitatively that reward design exerts its effect through a *trade-off dimension in risk profile* rather than a single Sharpe "alpha source." Second, we provide the first quantitative evidence that Kahneman-Tversky prospect theory confers out-of-distribution safety on RL trading policies. Code and data: https://github.com/cosmicpotato2047/capstone-rl-trading.

Keywords: Reinforcement Learning, Grid Trading, PPO, Reward Design, Prospect Theory, Differential Sharpe Ratio, Combinatorial Purged CV, Out-of-Distribution Generalization, Cryptocurrency

Contents

1 Introduction

Cryptocurrency markets exhibit micro-oscillations in hourly price that are largely orthogonal to long-term trends, motivating *grid trading* strategies that harvest revenue from volatility itself rather than from directional prediction. Grid trading is a simplified form of market making [Avellaneda and Stoikov, 2008], ranging from fixed-grid rules to volatility-adaptive variants that scale grid width by the Average True Range (ATR) [Wilder, 1978]. This work studies the latter: an ATR-proportional grid system whose grid width and limit-order placement are determined by a reinforcement-learning policy.

1.1 Motivation and Research Question

An earlier phase of this project hypothesized that PPO RL would discover dynamic volatility adaptation and outperform a fixed-grid baseline. Post-hoc behavioral analysis, however, revealed that the trained policy converged to a near-constant action $[1.0, 0.0]$ (§4), and that the Bayesian-optimized ATR coefficient itself was the principal source of performance. This negative finding suggested that the symmetric-reward + ATR-proportional combination structurally absorbs the policy's degrees of freedom, motivating the new research question of this paper:

> **In BTC grid trading, how does the design of the reward function affect the behavioral pattern and generalization performance of RL policies? In particular, do loss-asymmetric (asymmetric, prospect-theoretic) or risk-adjusted-return (DSR) reward formulations yield meaningful differences in single-metric advantage or out-of-sample robustness?**

1.2 Contributions

This work makes the following four contributions.

1. **A statistically rigorous comparison framework for reward variants.** Four reward functions (`sym`, `asym`, `dsr`, `pt`) are trained in an identical environment with 10 seeds × 1M steps, and pairwise statistical tests (Cohen's d, bootstrap P(A>B), IQM, 5% CVaR) are applied following Henderson et al. [2018]'s reproducibility guidelines.

2. **Scenario D — Pareto-frontier discovery.** Rather than a single "best variant" winner, the four variants statistically partition into two clusters {`sym`, `dsr`} vs. {`asym`, `pt`} forming a Pareto-like frontier on the Sharpe–MDD plane (§5). This outcome falls into none of the pre-registered scenario branches (A optimistic / B neutral / C pessimistic).

3. **Causal chain: reward → behavior → outcome.** Policy-level differences across variants are quantified on 1.04M step trajectories via a policy-distance matrix (within-cluster 0.129 vs. across-cluster 0.286), regime- specific trade/hold statistics, and action distribution comparison, confirming the mechanism reward loss asymmetry → reduced trade frequency → conservative cluster (§6).

4. **Winner reversal across evaluation environments and OOS robustness of prospect theory.** The winner reverses completely across single-split Val (`sym` 1$^{\text{st}}$) → CPCV multi-split

(`dsr` 1st) → the unsealed Test (`pt` 1st, consistent across two sources $p < 0.002$) (§7). The OOS robustness of `pt` is mechanistically attributed via trajectory analysis to its loss aversion $\lambda = 3.30$ and concave utility $\alpha = 0.68$ training policies to exit within mean 1.4 h, thereby avoiding the sell-side timing risk of the unseen bull regime. This is, to our knowledge, the first quantitative demonstration that the Kahneman-Tversky prospect theory [Kahneman and Tversky, 1979] confers OOS safety on RL trading policies.

1.3 Paper Structure

Section 2 surveys related work on RL trading, reward design theory, and evaluation methodology. Section 3 details the environment design (MDP, ATR formula, four reward variants) and PPO training. Section 4 briefly reproduces the Phase 1 negative finding ("RL = Fixed $[1.0, 0.0]$") as motivation. Section 5 reports the two-cluster Pareto-frontier result on Val (Scenario D), while Section 6 quantifies the mechanism. Section 7 presents the robustness trio: slippage (§7.1), CPCV multi-split (§7.2), and the central contribution of this paper—the Test OOS evaluation and the mechanistic basis of prospect theory's robustness (§7.3). Section 8 discusses distribution shift quantification and limitations, and Section 9 concludes.

2 Related Work

[TBD. Subsections mapped from PAPER_OUTLINE §2: market making and grid trading (Avellaneda-Stoikov 2008); DRL trading (Zhang-Zohren-Roberts 2020, Gort 2022, Sun 2023, Liu 2025); reward design theory (Moody-Saffell 2001 DSR, Kahneman-Tversky 1979 PT, Ng 1999 reward shaping); evaluation methodology (Lopez de Prado 2018 CPCV, 2014 DSR, Henderson 2018 reproducibility).]

3 Method

[TBD. Subsections: §3.1 MDP (State 7D, Action 2D, ATR-proportional formula); §3.2 trading-env mechanics (cycles, fee, slippage); §3.3 four reward variants + Optuna tuning (exp032a); §3.4 PPO stabilization (target_kl, ent annealing); §3.5 evaluation protocol (Val / CPCV / Test).]

4 Phase 1: Policy Saturation under Symmetric Reward

[TBD. Reproduces the "RL = Fixed $[1.0, 0.0]$" finding from exp020–022 of the earlier phase. Motivates reward formulation in this paper.]

5 Pareto Frontier and Scenario D

[TBD. exp032b results: 4 variants × 10 seeds, pairwise statistics, Pareto scatter, two-cluster discovery, hypothesis H1–H3 evaluation, Scenario D confirmation.]

6 Mechanism Quantification

[TBD. exp032c: policy distance matrix, behavior per regime, counterfactual action map, action distribution; causal chain reward → trade frequency → cluster.]

7 Robustness Analysis

7.1 Slippage Robustness

[TBD. exp033 results with slippage 0.02%. Cluster preservation ratio 2.19×; ATR-with-slippage fair-comparison results (Phase 16a).]

7.2 Multi-Split Evaluation via CPCV

[TBD. exp034: 6-fold CPCV 15 paths. DSR $p < 0.001$ *(Bonferroni 4-way correction);* `dsr` *reversal to 1st (1.413, 5% CVaR 0.890); cluster preservation ratio 3.55×.]*

7.3 Out-of-Sample Evaluation and Prospect Theory's Robustness

[TBD. Central chapter. exp035: 100 RL models + ATR baseline + Buy-and-Hold on Test. `pt` *1st on Test (both sources* $p < 0.002$*);* `dsr` *last on Test. Phase 16d mechanism: hold-duration distribution (DSR 7-day max vs. pt 6h max); causal account of* `pt`*'s OOS robustness.]*

8 Discussion

[TBD. Subsections: §8.1 Val–Test distribution-shift quantification (KS $p < 10^{-10}$, -27% *variance); §8.2 honest acknowledgment of environment dependence in exp027_rl prior evidence (Env-v3* `asym` *1.955); §8.3 limits of single-metric winner; §8.4 limitations: BTC-only, single regime in Test, single slippage level.]*

9 Conclusion

[TBD. Four main messages: (1) Pareto-frontier trade-off; (2) winner reversal across evaluation environments; (3) OOS robustness of Prospect Theory; (4) in-sample advantage ≠ OOS robustness. Future work: multi-asset, additional regimes, DR, meta-RL.]

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